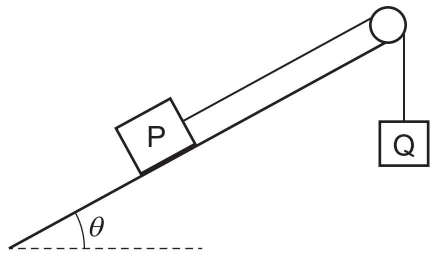
16 The diagram shows a block P of mass M connected by a string over a frictionless pulley to a block Q of mass m.



Block P moves up the slope with a constant velocity v. The slope is at angle θ to the horizontal.

The acceleration of free fall is g.

The resistive forces on the blocks are negligible.

Which expressions give the energy transferred per unit time to block P?

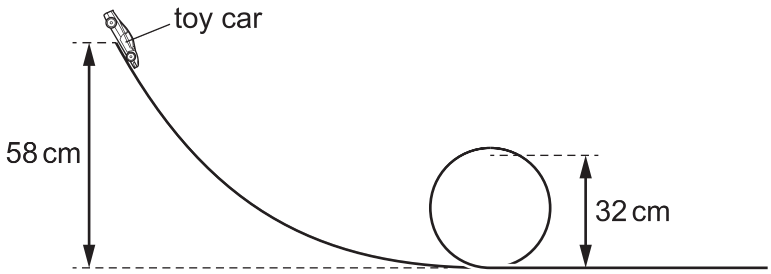
1 mgv

2 $Mgv \sin \theta$

3 $(M + m)gv$

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 2 only

17 A toy car travels around a vertical loop track.



The toy car is released from rest at a height of 58 cm above the bottom of the vertical loop.

The car is at a height of 32 cm when it is at the top of the vertical loop.

Assume that no resistive forces act on the car.

What is the speed of the car at the top of the vertical loop?

A $0.87\,m\,s^{-1}$ **B** $2.3\,m\,s^{-1}$ **C** $2.5\,m\,s^{-1}$ **D** $3.4\,m\,s^{-1}$